The Alan Turing Institute

# Containers (15:00-16:00)

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# What are containers and why use them?

- Containers are a technology that allow applications to be packaged and isolated with their entire runtime environment.
- They can give the appearance of root access, on a system that you may not actually have root access to.
- They provide isolation from other (potentially conflicting) software installations.
- They may aid reproducibility.
- They can simplify development as well as deployment of software.
- Isambard-AI doesn’t come with a lot of pre-installed software and containers are one way of adding your own.

# Docker vs podman vs apptainer

- **podman-hpc** and **apptainer (a.k.a. singularity)** available on Isambard-AI
- Why not docker?
  - Security and open-source considerations
  - Docker requires root permissions on the host system by default vs. apptainer and podman-hpc can be run as user processes
- OCI and compatibility
- Why apptainer?
  - Works easily across multi-node configurations
  - Host MPI integration
  - Apptainer images are saved as `.sif` files

# Container registries

- **Docker hub** (https://hub.docker.com/)
  - Widest range of available images.
  - You can create an account and push your own images.
- **Nvidia NGC containers** (https://catalog.ngc.nvidia.com/?tab=container)
  - PyTorch – has Python, PyTorch & CUDA pre-installed (see here for docs)
  - CUDA – has CUDA installed but you will need to install Python packages (see here for docs)
  - ⚠️ ***Isambard-AI is AARCH64/ARM64 and those platforms should be preferred over AMD64.***
- **GitHub container registry** (docs here)
  - To pull container images built with GitHub actions
  - e.g. https://github.com/alan-turing-institute/reginald/pkgs/container/reginald

# Pulling an image

We've already pulled some images, you won't need to pull any new images for this session

1. Check your system CUDA version:
   - `srun --gpus=1 --time=1:15:00 --reservation=Turing_Workshop --pty /bin/bash`
   - `nvidia-smi --version`
   - You can use any image with container CUDA version ≤ your system CUDA version
2. Download the image you'd like to run:
   - e.g. `apptainer pull docker://nvcr.io/nvidia/pytorch:24.12-py3`
   - e.g. `apptainer pull docker://nvidia/cuda:12.6.3-cudnn-devel-ubuntu20.04`
   - e.g. `apptainer pull docker://alpine` (you will need to install your own CUDA)
3. By default, this will save the image in the current working directory as a SIF file.

# Run commands interactively in a container

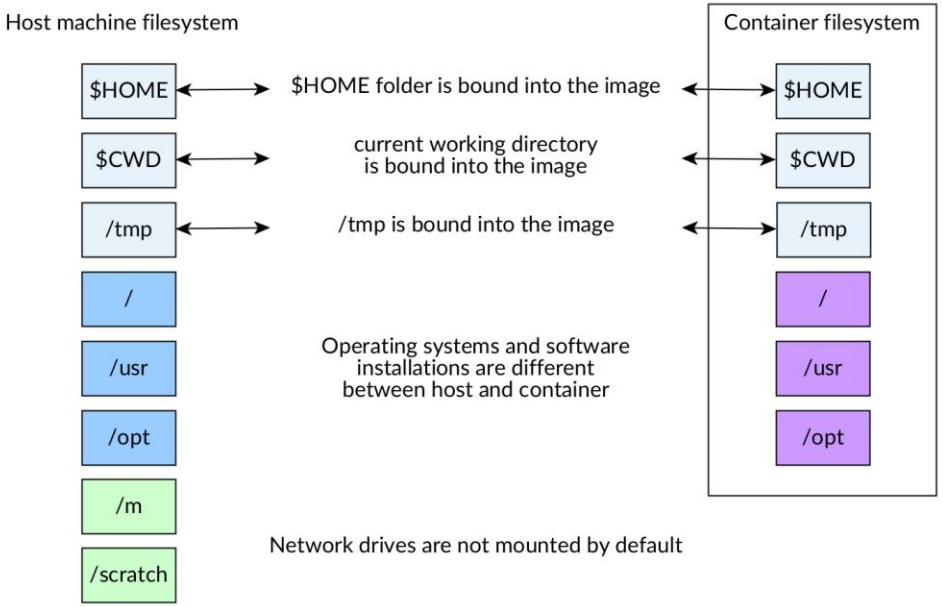
1. If you haven't already, start an interactive job:
   - `$ srun --gpus=1 --time=1:15:00 --reservation=Turing_Workshop --pty /bin/bash`
2. Change directory:
   - `$ mkdir $PROJECTDIR/$USER && cd $PROJECTDIR/$USER`
   - `$ cd -P .` (moves you to the real path, avoiding symlinks)
3. Start an interactive shell in your container:
   - `$ apptainer shell --nv $PROJECTDIR/pytorch_24.12-py3.sif`
   - We have pulled this image already and put it in the project folder
   - `--nv` tells apptainer to use NVIDIA GPUs
4. Run commands in the container, e.g. check you have torch installed with CUDA:
   - `Apptainer> python -c "import torch; print(torch.cuda.is_available())"`
   - ⚠️ In current set up, the container image is read-only and any python packages you install will be installed to ~/.local (i.e. your changes are local only)

# Ways of running commands in a container

1. Start an interactive shell in the container:
   - `apptainer shell [shell options...] <container.sif>`
2. Run a specific command in the container:
   - `apptainer exec [exec options...] <container.sif> <command>`
   - Best for batch scripts
   - `apptainer exec --nv pytorch_24.12-py3.sif python -c "import torch; print(torch.cuda.is_available())"` does the same as we did earlier but in a single command
3. Run a pre-defined default command within the container:
   - `apptainer run [run options...]<container.sif> [args...]`
   - Default command is defined when the image is built
   - Use `apptainer inspect -r` to see what it is.

# Container filesystems

- By default, apptainer mounts several host directories into the container filesystem.
  - user's home directory ($HOME)
  - current working directory ($PWD)
  - /tmp, /dev, /proc, /sys and some others
  - These mounted directories are writable
- We can tell apptainer not to mount with e.g. `--no-mount tmp,dev`
- Or we can tell apptainer to mount additional directories with `--bind src[:dst[:opts]][,...]`
  - e.g. `apptainer shell --bind /scratch --nv myimage.sif`
  - ⚠️ *Remember to manually mount user-specific tmp on Isambard-AI with* `--bind $TMPDIR:/tmp`
- Docker and podman have different behaviour (no default mounts).



# Installing packages in your container

1. By default, the container filesystem is read-only (except for the mounted directories), i.e. the following will error:
   - `$ apptainer shell $PROJECTDIR/alpine_latest.sif`
   - `Apptainer> cd / && touch x`
   - Also prevents installing packages into the container.

Three options for installing packages:

1. Use a venv located in a mounted directory (e.g. $CWD)
   - Avoid using your $HOME directory due to issues with storage quota
   - Use the container python to create your venv, then pip install into it
2. Use a sandbox container
3. Write your own def file

# Writeable filesystems /sandbox containers

You can interactively build your own container image using the `--sandbox` flag:

1. First, build a “sandbox” image from a base image:
   - `$ apptainer build --sandbox $TMPDIR/alpine $PROJECTDIR/alpine_latest.sif`
2. Edit it by, for example, installing software:
   - `$ apptainer shell --writable $TMPDIR/alpine`
   - `Apptainer> apk add python3`
   - `Apptainer> python3 --version`
3. And then turn the sandbox into .sif format with another build step:
   - `$ apptainer build alpine_python.sif $TMPDIR/alpine`

# Building a custom apptainer image with a .def file

1. Choose a base image:
   - NVIDIA CUDA and PyTorch images are solid choices for most Isambard-AI use cases
   - Basic images (ubuntu, alpine, etc) - you then install everything you need
2. Write your `.def` file
3. Build your image with:
   - `apptainer build container.sif container.def`
   - ⚠️ ***This should be done from a compute node***
4. Run your container image as before

container.def

```
Bootstrap: localimage
From: /path/to/pytorch_24.12-py3.sif

%post
  # install python packages
  pip install --no-cache-dir \
  transformers==4.48.3 \
  huggingface_hub[cli] \
  datasets \
  accelerate \
  tokenizers

%environment
  # set cache directory
  export HF_HOME=/tmp/huggingface

%runscript
  exec python "$@"
```

# Multi-node training with apptainer

- Same training code as for earlier lesson with BriCS team
- Isambard-AI requires multi-node specific setup steps:
  1. Add `module load brics/apptainer-multi-node` to your batch script
  2. Add `/host/adapt.sh` to the beginning of scripts/commands you will run in the container
- Isambard-AI docs here

sbatch_pytorch_container.sh

```
#!/bin/bash
#SBATCH --job-name=Torch_Distributed
#SBATCH --nodes=2
#SBATCH --gpus=8
#SBATCH --time=00:10:00
#SBATCH --ntasks-per-node=4

module load brics/apptainer-multi-node

export MASTER_ADDR=$(scontrol show hostname $SLURM_NODELIST
| head -n 1) # e.g. nid001038
export MASTER_PORT=29600

echo "Job Started at $(date)"

# Run the function with srun
srun -N 2 \
    --gpus=8 \
    --mpi=pmi2 \
    --ntasks-per-node=4 \
    bash -c 'export WORLD_SIZE=$SLURM_GPUS; export
RANK=$PMI_RANK; export LOCAL_RANK=$SLURM_LOCALID; apptainer
exec --nv container.sif bash -c "/host/adapt.sh; python3
train.py"'

echo "Job Finished at $(date)"
```

# Using a venv

Repeat what we've just done using the venv approach:

1. Start with the pytorch container image (`pytorch_24.12-py3.sif`)
2. Launch the container using `apptainer shell`
3. Create a venv using `python -m venv`
4. Activate your venv
5. Install the packages needed for running the train script
6. Adapt the batch script to use your venv python

# Build a container that behaves like a CLI tool

1. Find a useful tool, which isn't already available on Isambard-AI.
   - We suggest ripgrep (a faster grep) or azcopy (for transferring data from Azure)
2. Choose a base image.
   - Ubuntu and Alpine both have ripgrep available as packages.
3. Create a sandbox image, so that you can verify the install commands.
   - i.e. `apt-get install <package>` or `apk add <package>` or follow from docs
4. Write a definition file.
5. Build the image.
6. Make it executable with `chmod +x container.sif`
   - Optionally, make a symlink to it called "rg" with `ln -s /path/to/ripgrep.sif rg`.
7. Add the enclosing directory to your PATH environment variable.
8. Use it as you would any other command:
   - `rg -e "pattern" -r /path/to/some/directory`
   - Optionally, prepend `/usr/bin/time` to see how long it takes.